

Anthony Bardou

POSTDOCTORAL RESEARCHER IN MACHINE LEARNING

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Experience

École Polytechnique Fédérale de Lausanne

Lausanne, CH

POSTDOCTORAL RESEARCHER @ INDY LAB

Oct. 2023 - Now

- **Collaboration:** Prof. Patrick THIRAN
 - Researched theoretical guarantees and applications of Bayesian optimization algorithms in several contexts.
- Keywords:** Online Learning, Decentralized Algorithm, Time-Varying Optimization, Bayesian Optimization, PyTorch

École Normale Supérieure de Lyon

Lyon, FR

RESEARCH INTERN @ LIP/DANTE

Apr. 2020 - Sep. 2020

- **Subject:** Machine Learning for the Spatial Reuse of Wi-Fi Networks.
 - **Supervision:** Prof. Thomas BEGIN, Prof. Anthony BUSSON
 - Problem addressed from a centralized, reinforcement learning perspective, evaluated with an homemade Wi-Fi network simulator.
- Keywords:** Online Learning, Multi-Armed Bandit, Spatial Reuse

Université Claude Bernard Lyon I

Lyon, FR

RESEARCH INTERN @ LIRIS/SyCoSMA AND ELICO

Dec. 2018 - Jun. 2019

- **Subject:** Automatic Characterization of Citation Intentions in Scientific Publications.
 - **Supervision:** Prof. Frédéric ARMETTA, Prof. Marc BERTIN
 - Researched and contributed to attention and embedding techniques adapted to citations.
- Keywords:** Natural Language Processing, Embedding, Attention, Deep Learning, PyTorch

Université de Perpignan Via Domitia

Perpignan, FR

UNDERGRADUATE RESEARCH INTERN @ LIRMM/DALI

Apr. 2018 - Jun. 2018

- **Subject:** Study the Robustness of a Non Intrusive System Capable of Measuring the Human Respiratory Volume.
 - **Supervision:** Assistant Prof. David PARELLO, Prof. Henri MÉRIC
 - Produced a robustness analysis and performed a computational complexity reduction through combinatorial optimization.
- Keywords:** Combinatorial Optimization, Robustness Analysis, Image Processing

Education

École Normale Supérieure de Lyon

Lyon, FR

PHD IN MACHINE LEARNING AND WIRELESS NETWORKS

Oct. 2020 - Sep. 2023

- **Subject:** Online Learning for the Optimization of Wireless Networks and Beyond.
- **Supervision:** Prof. Thomas BEGIN
- **Award:** GDR RSD/ASF Best Ph.D. Thesis
- Researched zeroth-order optimization algorithms to optimize high-dimensional noisy black-box functions in a variety of applications, with a strong focus on wireless networks.

Université Claude Bernard Lyon I

Lyon, FR

M.S. IN DATA SCIENCE

Sep. 2018 - Sep. 2020

- Rank: 1 / 47
- Highest Honors

Université de Perpignan Via Domitia

Perpignan, FR

B.S. IN COMPUTER SCIENCE AND MATHEMATICS

Sep. 2015 - Jul. 2018

- Rank: 1 / 16
- Highest Honors

Skills

Data Science	Machine Learning, Data Mining, Big Data, Probabilistic Graphical Models
Applied Math.	Bayesian Inference, Optimization, Statistics, Performance Evaluation, Operations Research
Programming	PyTorch, BOTorch, Python, C/C++, Java, JavaScript
Languages	French, English, Spanish (notions), German (notions)

Honors & Awards

INTERNATIONAL

2025	Top Reviewer , 39th Annual Conference on Neural Information Processing Systems	San Diego, US
2024	Top Reviewer , 38th Annual Conference on Neural Information Processing Systems	Vancouver, CA
2022	Best Paper Award , 25th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems	Montreal, CA

NATIONAL

2025	Best Ph.D. Thesis , GDR Réseaux et Systèmes Distribués / Association ACM SIGOPS France	Toulouse, FR
2022	Best Student Paper Award , 24ème Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications	Paris, FR

Publications

INTERNATIONAL CONFERENCES

ANTHONY BARDOU, PATRICK THIRAN, GIOVANNI RANIERI. This Too Shall Pass: Removing Stale Observations in Dynamic Bayesian Optimization. *Proc. NeurIPS'24*. Vancouver, Canada. Dec. 2024

ANTHONY BARDOU, PATRICK THIRAN, THOMAS BEGIN. Relaxing the Additivity Constraints in Decentralized No-Regret High-Dimensional Bayesian Optimization. *Proc. ICLR'24*. Vienna, Austria. May 2024.

ANTHONY BARDOU, THOMAS BEGIN. INSPIRE: Distributed Bayesian Optimization for ImproviNg SPatial REuse in Dense WLANs. *Proc. MSWiM'22*. Montréal, Canada (QC). Oct. 2022. 10.1145/3551659.3559050. **BEST PAPER AWARD**.

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Improving the Spatial Reuse in IEEE 802.11ax WLANs: A Multi-Armed Bandit Approach. *Proc. MSWiM'21*. Alicante, Spain. Nov. 2021. 10.1145/3479239.3485715.

PEER-REVIEWED JOURNALS

ANTHONY BARDOU, JEAN-MARIE GORCE, THOMAS BEGIN. Assessing the Performance of NOMA in a Multi-Cell Context: A General Evaluation Framework. *IEEE Transactions on Wireless Communications*. Jul. 2025.

SAMIR SI-MOHAMMED, **ANTHONY BARDOU**, THOMAS BEGIN, ISABELLE GUÉRIN-LASSOUS, PASCALE VICAT-BLANC.. NS+ NDT: Smart Integration of Network Simulation in Network Digital Twin, Application to IoT Networks. *Future Generation Computer Systems*. Mar. 2024.

ANTHONY BARDOU, THOMAS BEGIN. Analysis of a decentralized Bayesian optimization algorithm for improving spatial reuse in dense WLANs. *Computer Communications, ACM MSWiM 2022 Special Issue*. Jun. 2023. 10.1016/j.comcom.2023.06.004

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Mitigating Starvation in Dense WLANs: A Multi-Armed Bandit Solution. *Ad Hoc Networks*, Vol. 138. Jan. 2023. 10.1016/j.adhoc.2022.103015

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Analysis of a Multi-Armed Bandit Solution for Improving the Spatial Reuse of New Generation WLANs. *Computer Communications, ACM MSWiM 2021 Special Issue, Pages 279-292*. Sep. 2022. 10.1016/j.comcom.2022.07.015

PH.D. THESIS

A. BARDOU. Online Learning for the Black-Box Optimization of Wireless Networks. *ENS Lyon*. Lyon, France. Sep. 2023. **GDR RSD/ASF BEST PH.D. THESIS AWARD**.

External Accolades

Prix de thèse du GDR RSD : des travaux sur l'optimisation récompensés	Jan. 2025	INS2I (CNRS)
Lauréat du prix de thèse GDR RSD - Édition 2024	Sep. 2024	GDR RSD (CNRS)
L'optimisation bayésienne au service des performances du WiFi	Apr. 2023	INS2I (CNRS)

Schools

Spring School of Theoretical Computer Science on Machine Learning (EPIT'22)	Luminy, France
CENTRE INTERNATIONAL DE RECONTRES MATHÉMATIQUES (CIRM)	May 2022

Talks

Sep. 2025	Invited Speaker , ICube Lab (Strasbourg University)	Strasbourg, FR
May 2025	Invited Speaker , Secondmind Research Seminar	Cambridge, UK
Mar. 2025	Invited Speaker , Journées GDR Réseaux et Systèmes Distribués	Toulouse, FR
Jun. 2024	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
May 2023	Invited Speaker , INDY Lab (EPFL)	Lausanne, CH
Oct. 2022	Oral Presentation , MSWiM'22	Montreal, CA
Jul. 2022	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
Jun. 2022	Oral Presentation , AlgoTel'22	Paris, FR
Feb. 2022	Oral Presentation , ROADEF'22	Lyon, FR
Dec. 2021	Invited Speaker , INRIA Ockham (ENS Lyon)	Lyon, FR
May 2021	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
Nov. 2020	Invited Speaker , Performance Evaluation for Networks	Lyon, FR
Jul. 2018	Invited Speaker , LIRMM (Perpignan University)	Perpignan, FR

Academic Service

REVIEWS

2025	International Conference on Learning Representations , International Conference
2025	Scientific Reports , Peer-Reviewed Journal (Nature)
2025	Transactions on Machine Learning Research , Peer-Reviewed Journal
2025	International Conference on Machine Learning , International Conference
2024	Computer Networks , Peer-Reviewed Journal (Elsevier)
2023	Neural Information Processing Systems , International Conference
2023	Computer Communication , Peer-Reviewed Journal (Elsevier)
2023	International Symposium on Information Theory , International Conference (IEEE)

TEACHINGS

Fall 2025	Stochastic Modeling , 30 hours @ EPFL (Bachelor)	Lausanne, CH
Fall 2024	Stochastic Modeling , 30 hours @ EPFL (Bachelor)	Lausanne, CH
Spring 2023	Preparation to the <i>agrégation</i> of Computer Science , 32 hours @ ENS Lyon (Master)	Lyon, FR
Spring 2022	Numerical Methods , 15 hours @ UCBL (Bachelor)	Lyon, FR
Spring 2022	Optimization , 15 hours @ UCBL (Bachelor)	Lyon, FR
Fall 2021	Introduction to Artificial Intelligence , 12 hours @ UCBL (Master)	Lyon, FR
Fall 2021	Optimization and Operations Research , 18 hours @ UCBL (Master)	Lyon, FR

SUPERVISION AND MENTORING

Spring 2025	Robin Jaccard , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Adrien Deschenaux , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2025	Aryan Ahadinia , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Antoine Merel , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Nahuel Plomb Veuve , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2024	Emanuele Mengoli , Master Thesis @ EPFL (from École Polytechnique)	Lausanne, CH
Spring 2024	Giovanni Ranieri , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2023	Hugo Martel , Bachelor Research Internship @ ENS Lyon	Lyon, FR

Open-Source Repositories

W-DBO: Wasserstein Distance-Based Dynamic Bayesian Optimization	May 2024	WDBO-ALGORITHM/wdbo_algo
DuMBO: High-Dimensional, No-Regret Bayesian Optimization	Feb. 2023	abardou/dumbo